

EE0827 Student Component Timeline (W01-W13)

Week-to-Lab Component Map

EE0827 Embedded System Applications

Week-to-Lab Component Map

Week	Lab focus	Components needed in that week	Notes
W01	Baseline measurement	Nucleo-F401RE, USB cable, jumper wires	Uses lab oscilloscope and multimeter
W02	Power input/regulation	AMS1117-3.3, 22uF caps, 100nF caps, 10uF cap, jumper wires	Breadboard is lab stock
W03	Signal conditioning/protection	1k/2k/3.3k/10k resistors, 100nF + 1uF caps	Lab tools: bench supply, function generator, scope, DMM
W04	Motor control/H-bridge	DC motor, logic-level MOSFET, motor driver (TB6612/L298N), 100uF electrolytic, 1k resistors	Flyback diode is lab stock
W05	Sensors and ADC	LM35, 10k resistor	LDR is lab stock
W06	RTOS foundations	2 LEDs, 2x330R resistors	Breadboard is lab stock
W07	RTOS communication	10k potentiometer	Potentiometer is lab stock
W08	I2C + IMU	2x4.7k pull-ups if needed	MPU6050 + logic analyzer listed as provided/shared

Week	Lab focus	Components needed in that week	Notes
W09	Midterm practical	Components from W01-W07	Exam pulls from earlier labs
W10	SPI + DMA	Either microSD module + card OR W25Q SPI flash module	Budget path: W25Q flash
W11	Robust communication	Second Nucleo or USB-UART adapter	Cheaper option: USB-UART adapter
W12	Reliability/watchdog	Flash or EEPROM for persistent logging	W25Q from W10 can be reused
W13	Debugging/jitter	Core kit: Nucleo + LEDs/resistors	Logic analyzer optional

Purchase Priority

1. Before W04: motor path parts (motor + MOSFET + driver).
2. Before W05: LM35.
3. Before W08: low-cost USB logic analyzer (if not guaranteed shared every session).
4. Before W10: W25Q SPI flash (recommended budget storage path).
5. Before W11: USB-UART adapter (optional but useful).